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SIEM SECURITY ANALYSIS REPORT

Generated: 2026-03-26 12:10:12 | Log Source Type: Postfix

Total Events	1000
Parsed Successfully	1000 (100%)
Risk Level	MEDIUM
ML Detected Threats	0

Overall Risk Score: 40/100 (MEDIUM)

1. EXECUTIVE SUMMARY

This comprehensive security report presents the analysis of **1,000** log events from the SIEM system. The analysis successfully parsed **1,000** events (100% success rate) with **0** events failing to parse.

■ **No ML-detected threats** were found in the analyzed log sample. The security posture appears normal.

2. AI-POWERED THREAT ANALYSIS

This section contains detailed threat analysis generated by our AI SOC Analyst.

SIEM Security Analysis Report

Date: 2024-02-29

Analyst: AI Security Analyst

Subject: Security Posture Analysis of Mail Server Logs

1. EXECUTIVE SUMMARY

The mail server logs indicate potential security concerns, primarily related to brute-force attacks, threat intelligence matches, and spam activity. While the overall log volume is dominated by informational entries, the presence of these alerts necessitates further investigation and remediation to prevent potential compromise and maintain the integrity of the mail infrastructure.

2. KEY FINDINGS

| Finding | Severity | Description |

|---|---|---|

| **Brute-Force Attacks (Multiple)** | HIGH | Multiple brute-force attack alerts suggest attackers are attempting to gain unauthorized access to mail accounts. This could lead to account compromise, data breaches, and further malicious activity. |

| **Threat Intelligence Match** | MEDIUM | A threat intelligence alert indicates that activity from a known malicious IP address or domain has been detected. This suggests a potential compromise or ongoing malicious campaign targeting the mail server. |

| **Spam Activity** | MEDIUM | Detection of spam activity suggests the mail server may be compromised or being used to relay spam. This can damage the organization's reputation and lead to blacklisting. |

| **High Connection Rate from Single IP** | MEDIUM | Log entries show a high connection rate from IP `45.115.119.123`, potentially indicating a denial-of-service (DoS) attempt or aggressive scanning activity. |

| **Large Email Size from Finance Account** | LOW | The finance account (`finance@tmv.edu.in`) sent an email of size 154720 bytes. While not inherently malicious, this large size warrants investigation to ensure it's legitimate and doesn't contain sensitive data or malware. |

3. ATTACK CHAIN ANALYSIS

Based on the alerts and log data, a potential attack chain can be inferred:

* **Reconnaissance (TA0043):** The high connection rate from `45.115.119.123` suggests potential scanning and reconnaissance to identify open ports and vulnerabilities.

* **Credential Access (TA0006):** The brute-force attack alerts indicate attempts to compromise user accounts through password guessing.

* **Initial Access (TA0001):** Successful brute-force attacks could grant attackers initial access to the mail server or user accounts.

* **Execution (TA0002):** If successful, attackers could use compromised accounts to send spam, phishing emails, or distribute malware.

* **Command and Control (TA0011):** The threat intelligence match suggests potential communication with a command and control server, indicating a more sophisticated attack.

This attack chain aligns with common email-based attack vectors, emphasizing the need for robust security measures to prevent each stage.

4. THREAT INTELLIGENCE

* **Suspicious IPs:**

* `45.115.119.123`: High connection rate, potentially associated with scanning or DoS attacks.

*** The IP address associated with the threat intelligence alert (not provided in the logs, but should be investigated based on the alert).**

* **Attacker TTPs (Tactics, Techniques, and Procedures):**

* **Brute-Force Attacks (T1110):** Attempting to guess passwords to gain unauthorized access.

* **Spamming (T1587.001):** Using compromised accounts or servers to send unsolicited emails.

*** Potential Command and Control (T1071):** Communication with known malicious IPs (based on threat intelligence).

5. RISK ASSESSMENT

Current Risk Level: MEDIUM

****Justification:**** The presence of brute-force attacks, a threat intelligence match, and spam activity indicates active threats targeting the mail server. While the number of error and warning logs is relatively low, the potential impact of a successful attack, including data breaches, reputational damage, and service disruption, warrants a medium risk assessment. The high connection rate from a single IP further elevates the risk.

6. RECOMMENDATIONS

1. ****Implement Multi-Factor Authentication (MFA) (M1026):**** Enforce MFA for all mail accounts to significantly reduce the risk of successful brute-force attacks. ****Priority: HIGH****

2. ****Investigate and Block Suspicious IPs:**** Thoroughly investigate the IP addresses `45.115.119.123` and the IP associated with the threat intelligence alert. Block these IPs at the firewall level to prevent further malicious activity. ****Priority: HIGH****

3. ****Strengthen Password Policies:**** Enforce strong password policies, including minimum length, complexity requirements, and regular password changes. ****Priority: MEDIUM****

4. ****Implement Rate Limiting:**** Configure rate limiting on the mail server to prevent excessive connection attempts from single IPs, mitigating potential DoS attacks. ****Priority: MEDIUM****

5. ****Review and Update Threat Intelligence Feeds:**** Ensure the SIEM and other security tools are using up-to-date threat intelligence feeds to detect and block known malicious actors. ****Priority: MEDIUM****

6. ****Analyze Email Content:**** Implement content filtering and malware scanning to detect and block spam, phishing emails, and malicious attachments. Investigate the large email sent from the finance account. ****Priority: LOW****

7. ****Monitor for Account Anomalies:**** Implement monitoring rules to detect unusual login activity, such as logins from unusual locations or at unusual times. ****Priority: LOW****

7. INCIDENT RESPONSE

1. ****Isolate Affected Accounts:**** If any accounts are suspected of being compromised, immediately disable them to prevent further malicious activity.

2. ****Conduct Forensic Analysis:**** Investigate the compromised accounts to determine the extent of the damage and identify any data that may have been accessed or exfiltrated.

3. ****Reset Passwords:**** Force password resets for all users, especially those who may have been targeted by the brute-force attacks.

4. ****Review Mail Server Configuration:**** Review the mail server configuration for any misconfigurations or vulnerabilities that may have been exploited.

5. ****Notify Relevant Stakeholders:**** Inform relevant stakeholders, including IT staff, management, and legal counsel, about the incident and the steps being taken to address it.
6. ****Document the Incident:**** Thoroughly document the incident, including the timeline of events, the actions taken, and the lessons learned.
7. ****Implement Lessons Learned:**** Use the lessons learned from the incident to improve security policies, procedures, and technologies.

This report provides a preliminary analysis of the mail server logs. Further investigation and analysis may be required to fully understand the scope and impact of the detected threats. Continuous monitoring and proactive security measures are essential to protect the mail infrastructure from future attacks.

3. THREAT INTELLIGENCE & IOCs

Top Source IP Addresses (by event count)

Rank	IP Address	Events	Risk Assessment
1	127.0.0.1	245	HIGH
2	209.85.210.52	16	LOW
3	209.85.161.69	16	LOW
4	45.115.119.123	11	LOW
5	76.223.181.129	11	LOW
6	45.115.119.124	9	LOW
7	40.107.51.105	9	LOW
8	185.93.89.95	9	LOW
9	192.168.1.254	8	LOW
10	209.85.218.53	8	LOW

Most Targeted User Accounts

Rank	Username	Events	Priority
1	iqac@tmv.edu.in	18	MEDIUM
2	infocenter@tmv.edu.in	14	MEDIUM
3	finance@tmv.edu.in	11	MEDIUM
4	0.001	10	MEDIUM
5	examination@tmv.edu.in	10	MEDIUM
6	kulguru@tmv.edu.in	8	MEDIUM
7	law@tmv.edu.in	8	MEDIUM
8	administration@tmv.edu.in	8	MEDIUM
9	phd@tmv.edu.in	6	MEDIUM
10	tmvcompscience@gmail.com	6	MEDIUM

5. SECURITY ALERTS

Total alerts generated: 4. Showing top 15 alerts.

Alert #1: bruteforce

Timestamp: 1900-03-18T16:11:13 | **Source:** N/A

Message: Attack detected: bruteforce (confidence: 85%)

Alert #2: threat_intel

Timestamp: 1900-03-18T16:28:39 | **Source:** N/A

Message: Attack detected: threat_intel (confidence: 80%)

Alert #3: spam_activity

Timestamp: 1900-03-18T16:29:07 | **Source:** N/A

Message: Attack detected: spam_activity (confidence: 60%)

Alert #4: bruteforce

Timestamp: 1900-03-18T16:34:43 | **Source:** N/A

Message: Attack detected: bruteforce (confidence: 85%)

6. LOG ANALYTICS & STATISTICS

6.1 Severity Distribution

Severity Level	Event Count	Percentage
CRITICAL	0	0.0%
ERROR	6	0.6%
WARNING	7	0.7%
INFO	987	98.7%
DEBUG	0	0.0%

6.2 Log Type Distribution

Log Type	Count
postfix	1000

6.3 Top Source IP Addresses

Rank	IP Address	Event Count	Activity Level
1	127.0.0.1	245	HIGH
2	209.85.210.52	16	LOW
3	209.85.161.69	16	LOW
4	45.115.119.123	11	LOW
5	76.223.181.129	11	LOW
6	45.115.119.124	9	LOW
7	40.107.51.105	9	LOW
8	185.93.89.95	9	LOW
9	192.168.1.254	8	LOW
10	209.85.218.53	8	LOW

7. RECOMMENDATIONS & REMEDIATION

◆ Critical

Immediately investigate all critical and high-severity alerts. Isolate affected systems if compromise is suspected.

◆ High Priority

Review and block suspicious IP addresses with high event counts. Implement additional monitoring on targeted accounts.

◆ Medium Priority

Enable enhanced logging for critical systems. Review access controls for accounts with failed authentication attempts.

◆ Best Practices

Ensure all systems are patched and up-to-date. Implement MFA for all privileged accounts. Regular review of security logs.

◆ Monitoring

Set up real-time alerts for threshold-based detections. Create correlation rules for multi-stage attack patterns.

◆ Response

Document and test incident response procedures. Ensure backup and recovery mechanisms are functional.

8. APPENDIX

Report Metadata

Report Generated	2026-03-26 12:10:12
Log Source Type	Postfix
Total Lines	0
Parsed Lines	0
Failed Lines	0
ML Attacks Detected	0
Total Alerts	4
Analysis Engine	FreeKhana SIEM v2.0 + AI SOC Analyst

Disclaimer: This report is generated automatically by FreeKhana SIEM system. The analysis is based on log data and machine learning models. Human review is recommended for critical security decisions.